ModelIt!

Student Activity Pack

Why Do Roller Coasters Go So Fast?

The Amazing Science of Energy and Speed

Grade 4 | NGSS 4-PS3-1 | Energy and Matter

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Have you ever been on a roller coaster? What was the scariest part?**

2. **Why do you think roller coasters go really fast at the bottom of a big hill?**

3. **Draw what you think happens to a roller coaster's speed as it goes up and down hills.**

4. **What do you think 'energy' means in science?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Hill Height | Speed at Bottom | Energy of Motion

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Small Hill

Hill Height set to low

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Giant Hill

Hill Height set to maximum

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Medium Hill

Hill Height set to 50%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Friction — The rubbing force between the coaster and the track that slows it down and turns energy into heat

Coaster Mass — How heavy the roller coaster car is — heavier cars have more energy at the same speed

Track Curve Shape — Whether the track is straight, curved, or looping affects how speed changes

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does a roller coaster go super fast at the bottom of a hill without any engine pushing it? Use evidence from your simulation to support your answers.

1. **Why does a roller coaster go fastest at the very bottom of the tallest hill?**

2. **If energy can't be created or destroyed, where does the roller coaster's energy come from in the first place?**

3. **Why does a roller coaster eventually slow down and stop if energy is never destroyed?**

4. **How is rolling a ball down a ramp similar to a roller coaster going down a hill?**

5. **How did your model help you understand something you can see at an amusement park?**

8 STEM Challenge: Design the Ultimate Marble Run

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YOUR ENGINEERING MISSION

Design a marble run track that makes a marble go as fast as possible at the bottom using what you learned about height and energy.

THE SCENARIO

A toy company wants to create the most exciting marble run ever! Your engineering team has been hired to design a track that uses hills and ramps to make the marble reach maximum speed. Use evidence from your model to make your design choices.

Engineering Design Thinking

1. WHAT IS THE TALLEST HILL YOU CAN BUILD WITH THE MATERIALS YOU HAVE?
2. HOW WILL YOU MEASURE THE SPEED OF YOUR MARBLE AT THE BOTTOM?
3. WHAT SHAPE SHOULD YOUR RAMP BE — STRAIGHT, CURVED, OR BUMPY?
4. HOW WILL YOU COMPARE DIFFERENT HILL HEIGHTS TO SEE WHICH GIVES THE MOST SPEED?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Roller Coaster Engineers and Mechanical Engineers design theme park rides by using math and science to make sure rides are thrilling AND safe. They earn $65,000–$110,000/year.